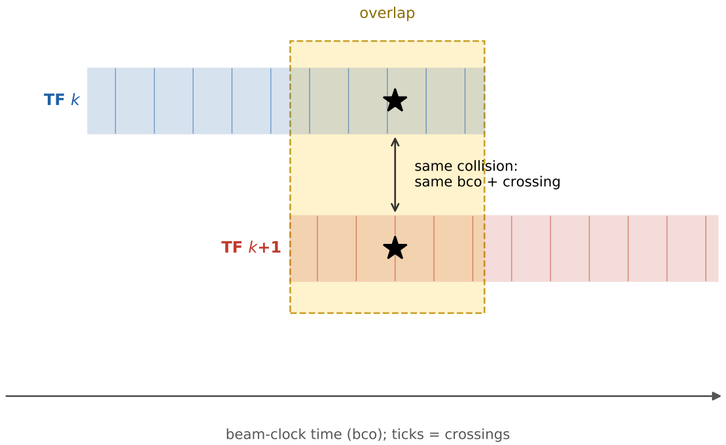
ana578, page 1: the clean sample of events -- whole-event copies, track copies across crossings, tracks from another collision -- schematic, not to scale

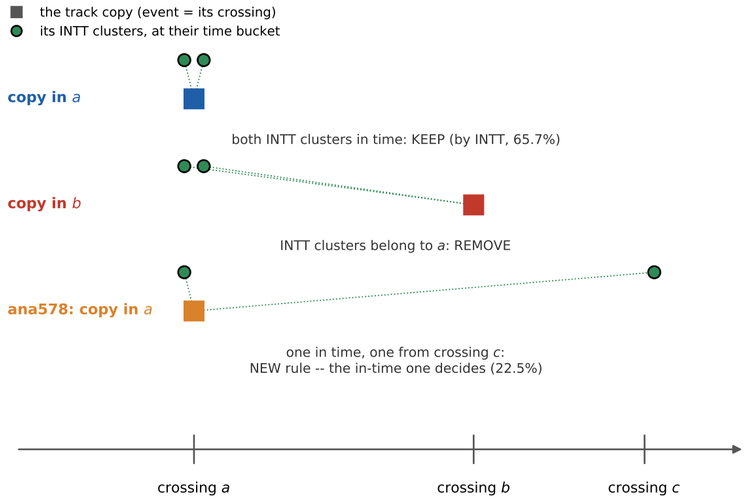
1. One collision in two overlapping trigger frames
TFdup: skip the second copy (whole event)



The copy in TF k+1 is the SAME event. Kept, it would sit in the mixing pool next to its original and mix with its own copy: a like-sign MIXED spike at (dy, dϕ) = (0,0), x7.97 with nothing on (ana573 x8.75).

TFdup skips 12.8% of the ana578 events (ana573 12.9%), 93% of them one TF later. First copy read is kept. With TFdup: x1.19; with XTF and the mixing exclusion too: 1.01.

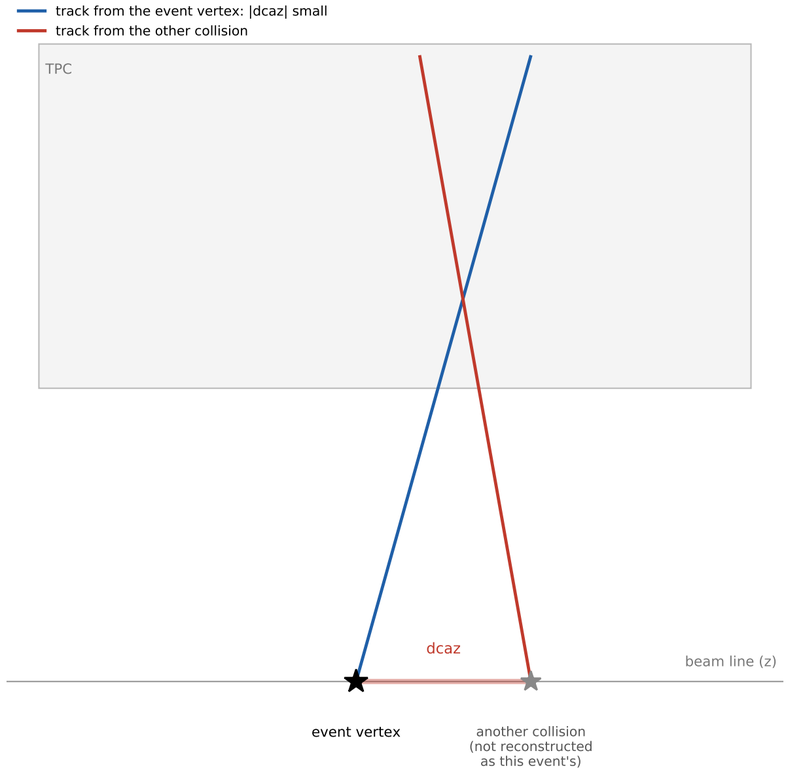
2. One track copied into another crossing of the same TF
XTF: keep the copy whose INTT clusters are in time



The matcher can attach one silicon seed (same Si cluster keys) to TPC tracks in different crossings of one TF: the same particle twice.

ana578: 23% of the tracks with $\geq$ 2 INTT clusters carry one from an unrelated crossing (ana573 0.9%, all within ±1). The old rule (all INTT clusters agree) then fell back to quality; judged by |dcaz| the in-time rule is right 0.941 of the time, quality 0.910.
158M duplicate pairs: INTT 65.7%, in-time 22.5%, quality 7.1%, neither in time 4.8% (both dropped); 3.0% of the tracks removed.

3. A track attached to the wrong vertex
level-0 cut |dcaz| $\leq$ 1.5 cm (side view, r vs z)

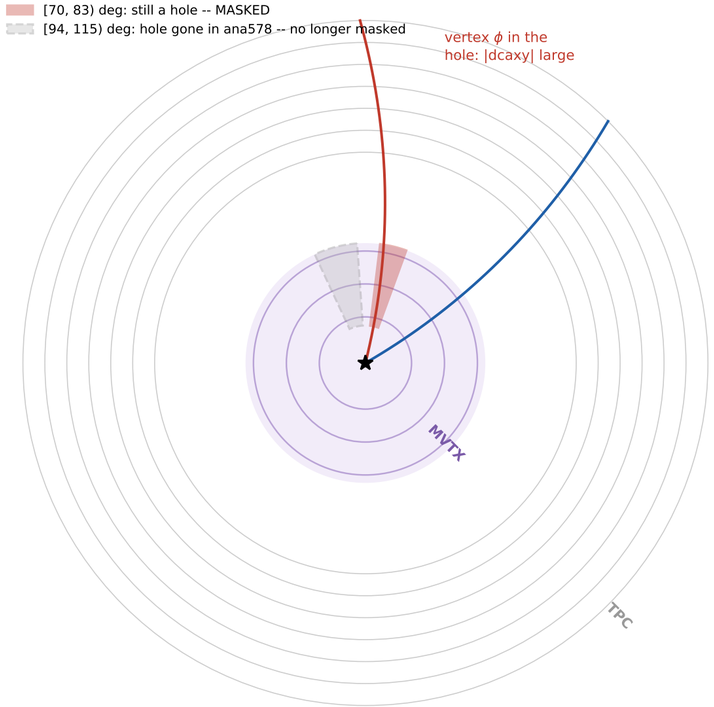


Raw XingView: jet- and photon-triggered TFs have more tracks per event near the trigger (|crossing| ~ 30; jet - MB +0.085 ± 0.017 at crossings -20..-5).
N-1 test per cut (20M rows of 79516): the event cuts do not touch the bump; with every track cut but dcaz it stays (+0.085), with dcaz it is gone (-0.010 ± 0.015).
So the bump is tracks with |dcaz| > 1.5 cm: from other collisions near the trigger, removed by the standard level-0 cut. All crossings above 2 are usable.

Event = one crossing with a vertex; TF = trigger frame (the block of streaming data that holds many crossings). Numbers: 6-run production ana578 (TFdup, XTF counts) and run 79516 (README_SplitTracks578 sec 1).

ana578, page 2: Area A, single-track acceptance -- holes that change from event to event, masked per track -- schematic and data

4. A1 -- the SiPhi zones: silicon holes in vertex ϕ mask [70, 83) deg only (SiPhiMask)



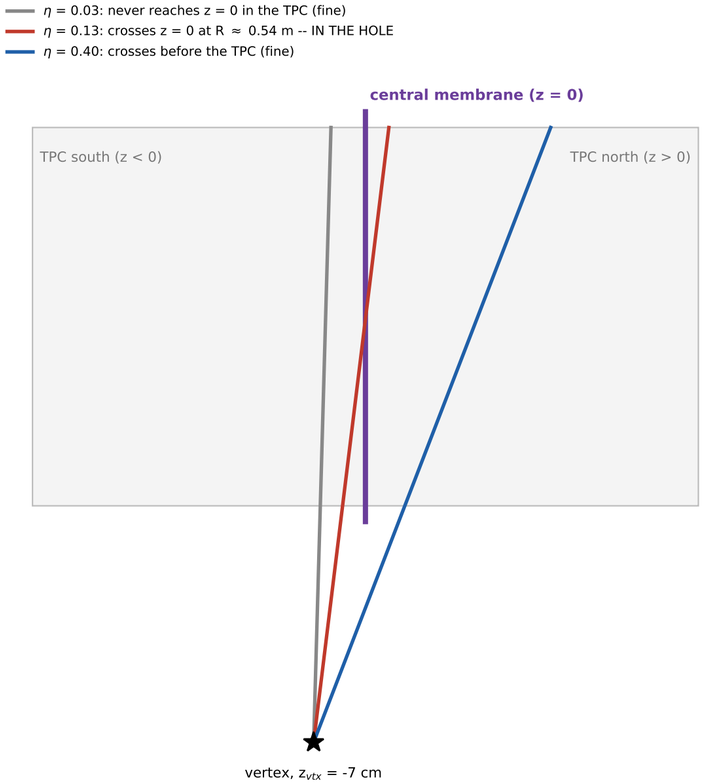
Tracks per 1-deg bin of vertex ϕ and their <|dcaxy|> (level-0 tracks, SiPhi mask off; ana573_795xx_ZB vs ana578_ZB, 1-2.5e9 tracks per charge):

[70,83): 2.2-2.3% of the tracks (flat 3.6%), <|dcaxy|> 1.6x the median (ana573 1.7-1.9%, 1.7-1.8x): sharp in vertex ϕ, both charges, a defect at the silicon radius -- shallower, but still there.

[94,115): 5.6-6.0% (flat 5.8%), <|dcaxy|> normal (ana573 2.5-2.7%, 1.4x): gone.

The mask costs 2.3% of the ana578 tracks (ana573 4.36% with both zones).

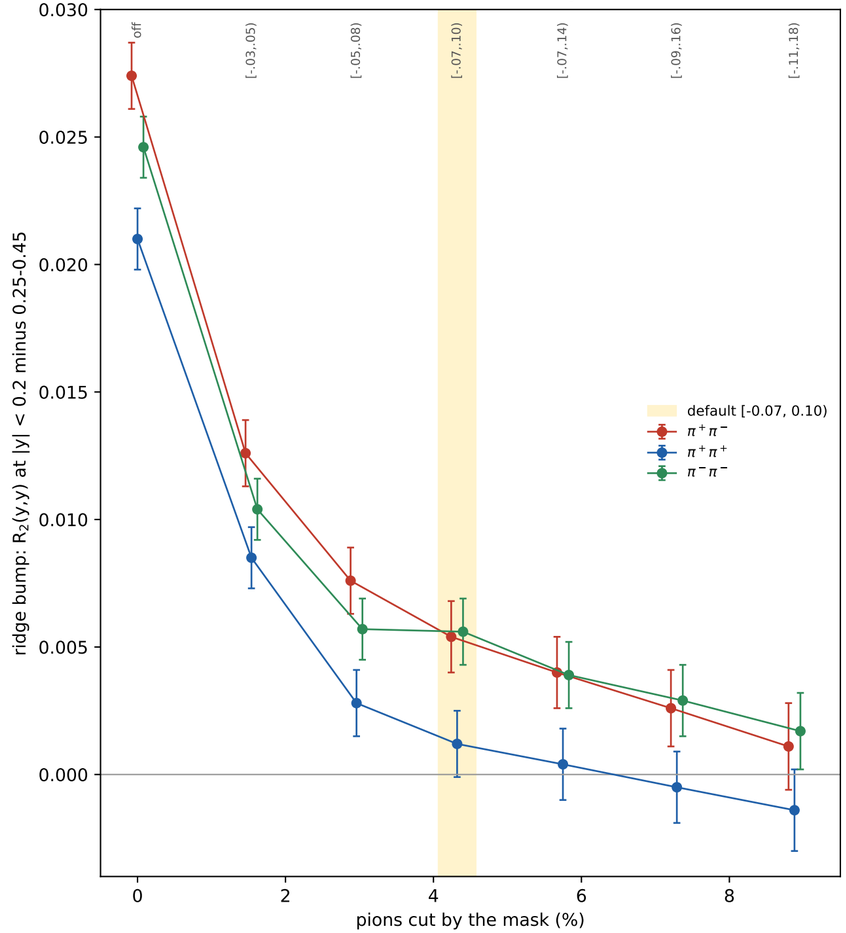
5. A2 -- the TPC central-membrane hole tracks that cross z = 0 inside the TPC (CM mask)



Side view (z stretched x5): a track crosses z = 0 at R = -z_{vtx}/sinh η; for R in the TPC its clusters sit on both sides of the membrane: a band in η that moves with z_{vtx}. w = (η + 0.018 z_c) sign(-z_c), z_c = the z_{vtx}-slice center: the hole sits at w $\approx$ 0.

ana578: same place and shape, half as deep (minimum 0.42 of the plateau; ana573 0.23). A hole that is the same in every event cancels in S/M; this one opens and shuts from event to event, so it makes a bump in $R_2(y_1,y_2)$ at $y_1 = y_2 \sim 0$.
Mask: w in [-0.07, 0.10), fixed per 2-cm slice: 4.32% of the pions (ana573 2.79%).

6. A2 decided after Area B: the bump vs the mask width all of 79516 (~280M events), every B cleaner on

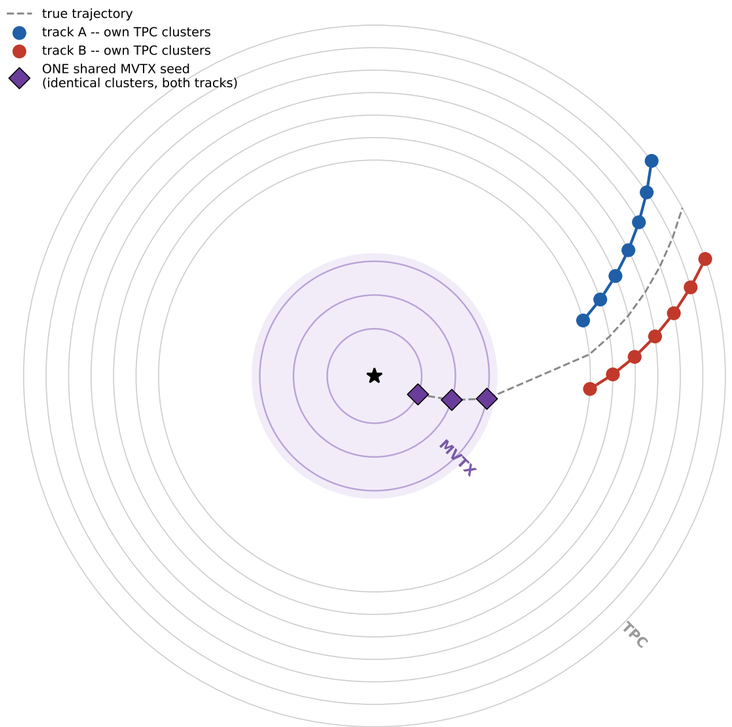


Without the mask a CHARGE-INDEPENDENT bump of 0.021-0.027 (16-21σ) is left after Area B; the mask removes it, monotonically with the width. It saturates near the default: [-0.05,0.08) -> [-0.07,0.10) gains 0.000-0.002 for 1.4% more pions; wider windows ~0.001-0.0015 per 1.5% of the pions, ~1σ each. Decision: keep [-0.07, 0.10).
$\pi^-\pi^-$ stays ~0.003 above $\pi^+\pi^+$ at every width, also with no mask: not the CM hole; a watch item (4σ broad offset on the $y_1 = y_2$ diagonal).

Numbers: README_SplitTracks578 secs A1 (SiPhi_ana578.pdf), A2 and the A2 width scan (CMMask_ana578.C, ST578_A2_CMMask.pdf); stat errors from the same-event bins (the widths are nested, same events, so neighbouring points are correlated).

ana578, page 3: Area B, like-sign split tracks -- one silicon seed inside two same-charge tracks -- schematic and data

7. Like-sign "duplicate"
in the pregate: SKF ≥ 0.10 AND SL < 0.18 (path 1), or ≥ 2 MVTX (path 3)



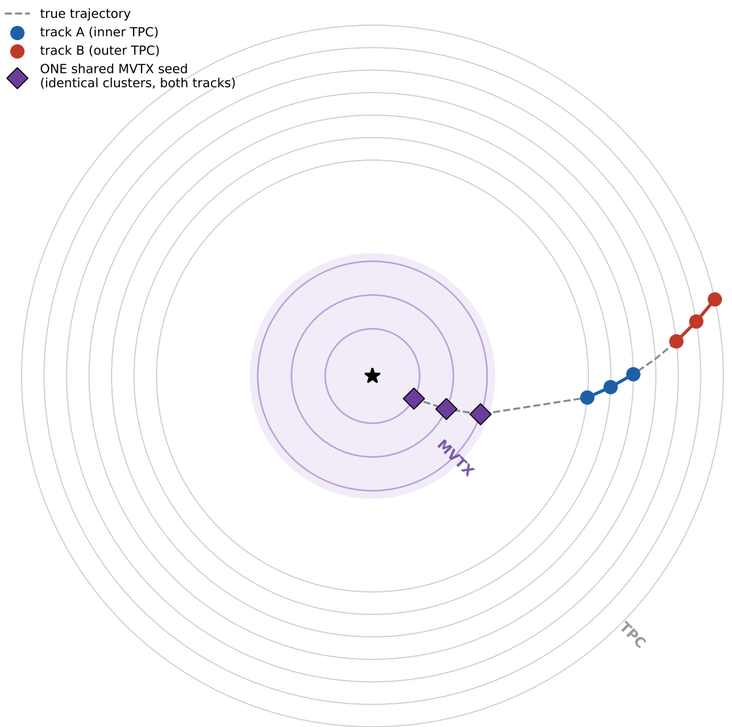
Same charge, same silicon seed, two TPC cluster sets on the SAME layers (different clusters). SL ≈ -0.45, RadialGap ≈ 0.02.

The fragments share the seed but can differ in p_T: they land at $(\Delta\eta,\Delta\phi) \approx (0, \Delta\phi_0)$; the pregate is centered there (panel 9).

ana578 (10M of 79516, B1): SL does 92% of the LS removals.
Path 3 (≥ 2 MVTX layers with the same cluster) is NEEDED: without it 33.6k more tracks stay and the like-sign bump comes back (+0.050 $\pi^+\pi^+$, +0.022 $\pi^-\pi^-$).

π^+ splits 1.80 ± 0.02 times as often as π^- (ana573 1.90).
LS paths, 6 runs: 1.77% of the accepted pions (ana573 1.74% of the tracks).

8. Like-sign "complementary"
in the pregate: SKF ≥ 0.10 AND RadialGap ≥ 0.45 (path 2)

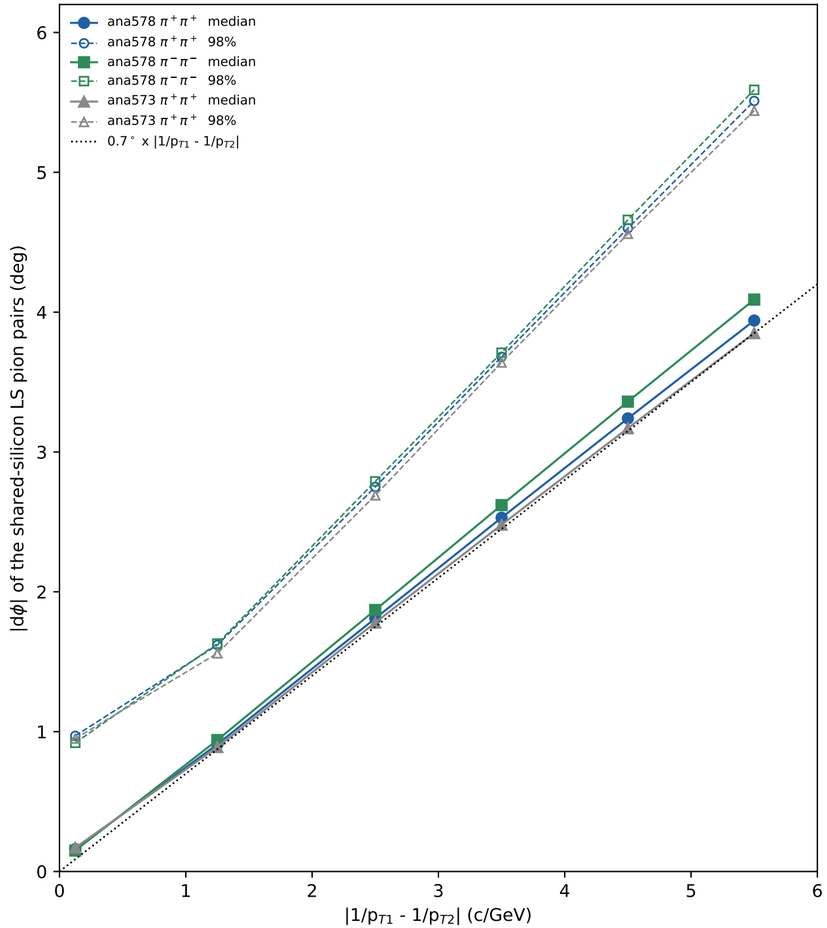


Same charge, same silicon seed, one trajectory's TPC clusters torn into DISJOINT inner/outer segments. SL = +1 (blind), RadialGap ≈ 0.53.

ana578: the RadialGap path removes 3-4x more than in ana573 (0.024% of the pions vs 0.006% of the tracks), 2.7x more π^- than π^+ -- but with RG off path 3 takes 7,537 of its 7,600 pairs: only which path gets the pair first.

Nearly redundant; kept (costs nothing, catches 63 pairs path 3 misses).

9. B1 -- the like-sign split band and the pregate
shared-silicon LS pion pairs, no cleaner (10M of 79516)

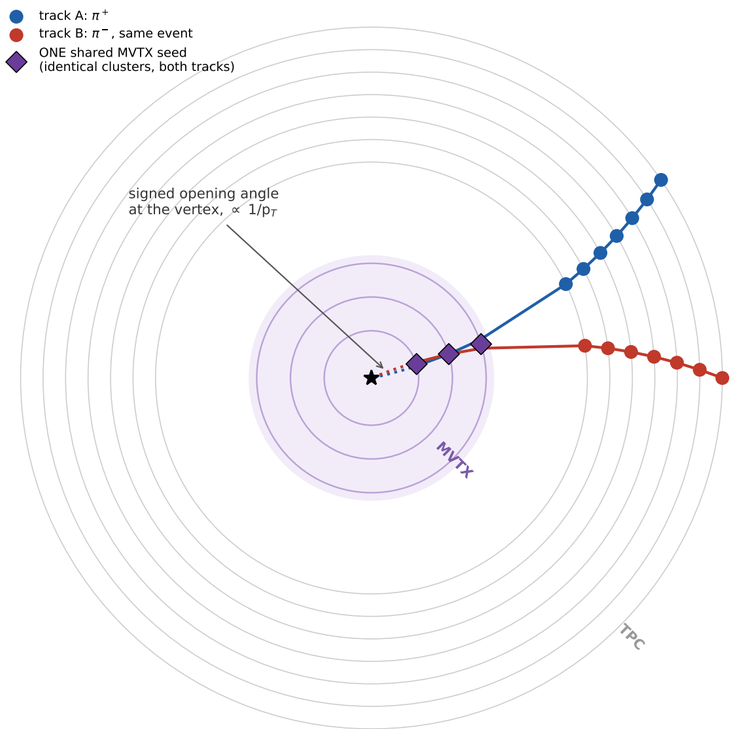


Same law as ana573: $|d\phi| \sim 0.7^\circ \times |1/p_{T1} - 1/p_{T2}|$, both charges; ana578 medians 0.03-0.10° higher (half a 0.25° bin or less); π^- ~0.1° above π^+ at large curvature difference. In the gate frame the core sigma is 0.13-0.34°, well inside the half-width w = 2.0-2.6°: 99.6-99.7% of the shared-silicon pairs are inside the pregate. Shared silicon off the band: ~0.3 pairs/deg vs 500-3000/deg on it.
Decision: the pregate stays (dps20, R = 6 cm, dη 0.040).

Purple diamonds = MVTX clusters; grey circles = TPC (schematic). The proof of "one silicon seed in two tracks" is Si cluster IDENTITY (same physical cluster). Loser rule: fewer TPC clusters, then worse fit quality. Numbers: README_SplitTracks578 secs B1-B3.

ana578, page 4: Area B, opposite-sign -- one silicon seed in two opposite-charge tracks, and loopers -- schematic and data

10. B4 -- one silicon seed, two opposite-charge TPC tracks
ULS veto: SiSplitScore ≥ 0.05, any angle



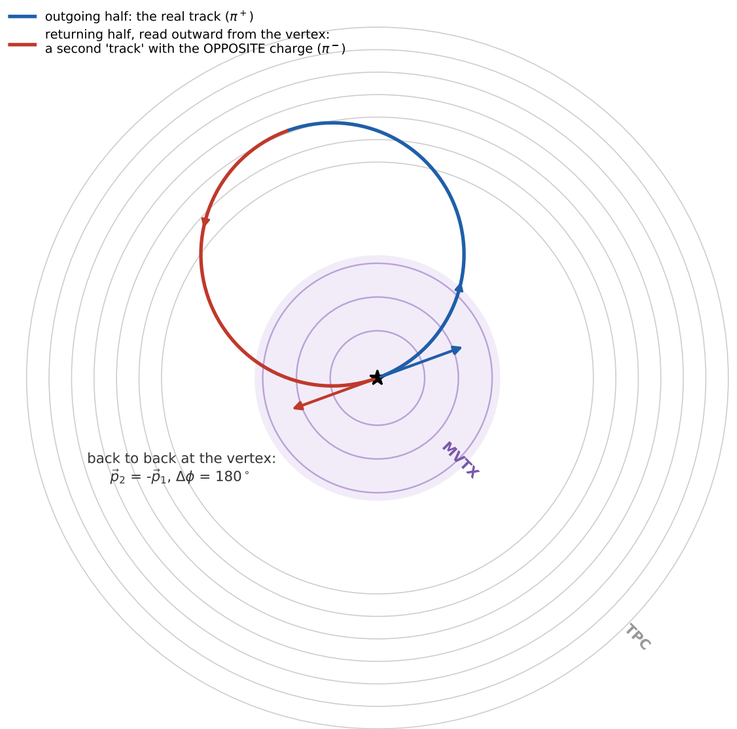
The matcher attaches one silicon seed to two TPC seeds of opposite curvature in the SAME crossing (other crossings of the TF: XTF, page 1).

A $\pi^+\pi^-$ pair at very small Q: the C(Q)/(1+r_2) bump at 20-35 MeV (panel 12) and the unlike-sign ridge bump at y ~ 0 (0.215 -> 0.030 with the veto).

ana578 (10M of 79516): 369,769 pairs flagged; losers π^+ 1.22%, π^- 1.09% of the accepted pions (ana573 1.10% of the tracks). Like-sign C(Q) unchanged.

Decision: the veto stays as it is.

11. B5 -- a looper: one curling track seen twice
looper veto: remove the worse half



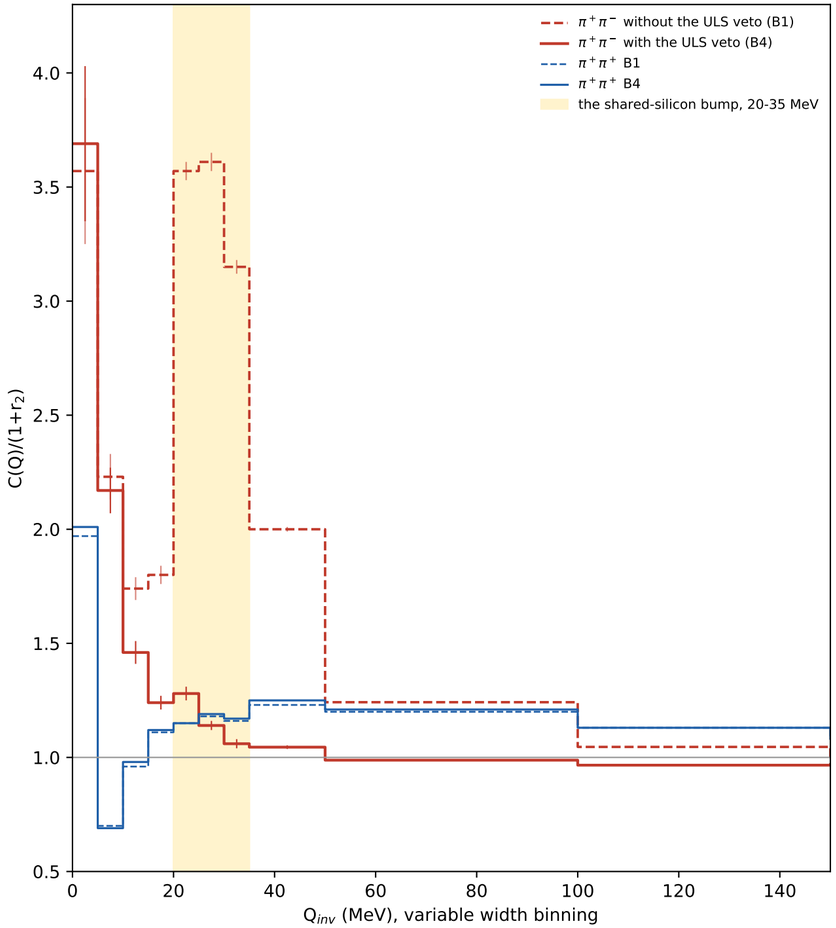
p_T < ~0.17 GeV/c: the circle closes inside the TPC and the track comes back through the center: an opposite-charge 'track', back to back, same |p|.

Veto: OS, both |p| < 0.20 GeV/c, $\Delta\phi \geq 170^\circ$, $|\vec{p}_1+\vec{p}_2|/(|p_1|+|p_2|) < 0.08$.

ana578 (all of 79516): 74.6k excess back-to-back pairs below 0.08 over a control of 28.7k (relative sum 0-0.02: 36x the like-sign control); the veto takes them to 0.91-0.95. ~20% of the excess lies above 0.08 (1.33 at 0.08-0.10): watch.

0.015% of the pions, 0.117% at p_T 0.1-0.2 (ana573 0.010%, 0.08%). $\pi^+\pi^-$ away-side spike at dy ~ 0: +0.030 -> +0.009. Decision: 0.08 stays.

12. B4 in the data: $\pi^+\pi^-$ C(Q)/(1+r_2)
10M of 79516, without and with the ULS veto

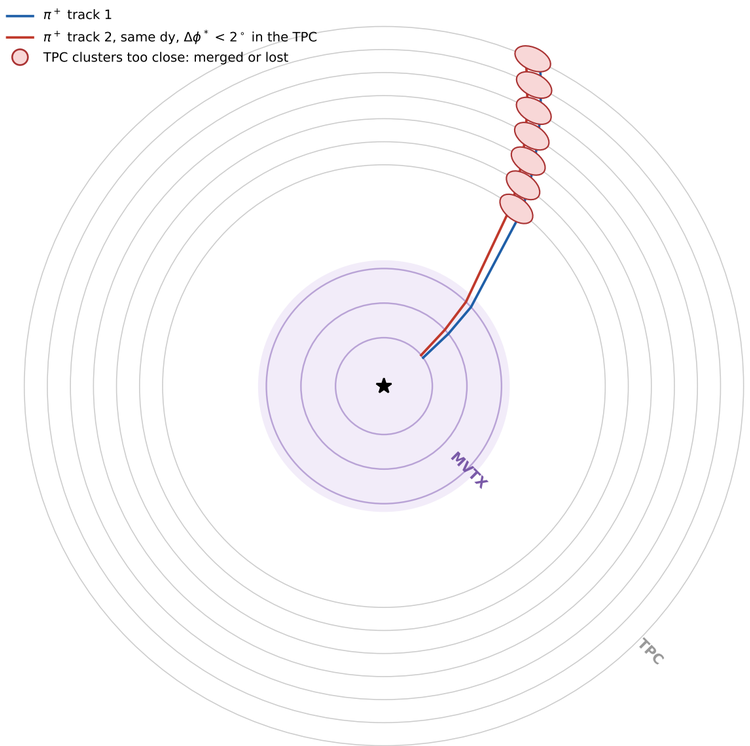


Without the veto: 3.6 at 20-35 MeV, on top of the rising Coulomb part (3.7, 2.2, 1.5 at 0-15 MeV). The veto removes the bump and leaves the Coulomb rise; the removed pairs reach to Q ~ 0.2 GeV (50-100 MeV: 1.24 -> 0.99). Like-sign within 0.01-0.02. 1+r_2 = total sibling / total mixed pairs (1.307 B1, 1.295 B4). "variable width binning".

Panel 10: the same-event case (ULS veto); the cross-crossing case is XTF (page 1, panel 2). Panel 11 is one particle (loser: fewer TPC clusters, then the larger |dca| sum). Numbers: README_SplitTracks578 secs B4, B5.

ana578, page 5: Areas C and D -- two real tracks, merged or lost (pair cuts), and the notch (watched) -- schematic, not to scale

13. C1 -- two like-sign tracks close all through the TPC
tsep: drop |dy| < 0.06 with min |dϕ^*| < 2.0° over R = 0.30-0.78 m

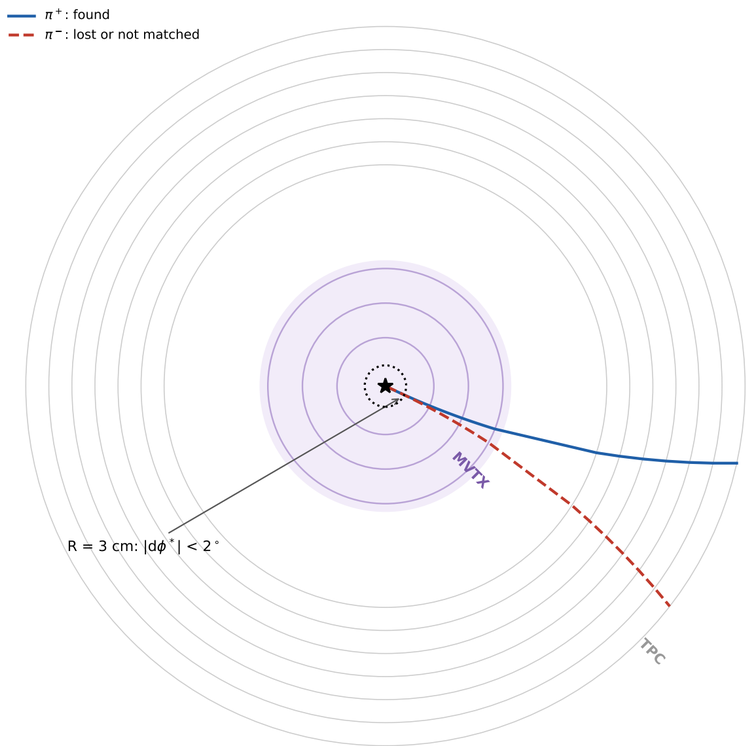


Two real tracks that stay within ~2° of each other through the TPC share or merge clusters: one is lost or both are damaged. Like-sign pairs at small Q do this (same curvature); unlike-sign pairs bend apart.

ana578 (all of 79516, tsep off): S/M 0.84-0.87 at min |dϕ^*| < 1°, |dy| < 0.04 (ana573 0.46-0.74). With tsep the LS C(Q)/(1+r_2) reaches its plateau (~1.30) from 20 MeV (off: still rising at 40 MeV); tsepm0630 gains < 0.01.

Cut pairs from S and M alike: LS pions 0.36-0.38% S / 0.36% M (6 runs). Open: a narrow S/M dip (0.92-0.94, < 1°) at |dy| 0.10-0.14, beyond any window.

14. C2 -- two tracks that leave the vertex together
isep: drop |dy| < 1.0 with |dϕ^*(R = 3 cm)| < 2.0°, LS and ULS

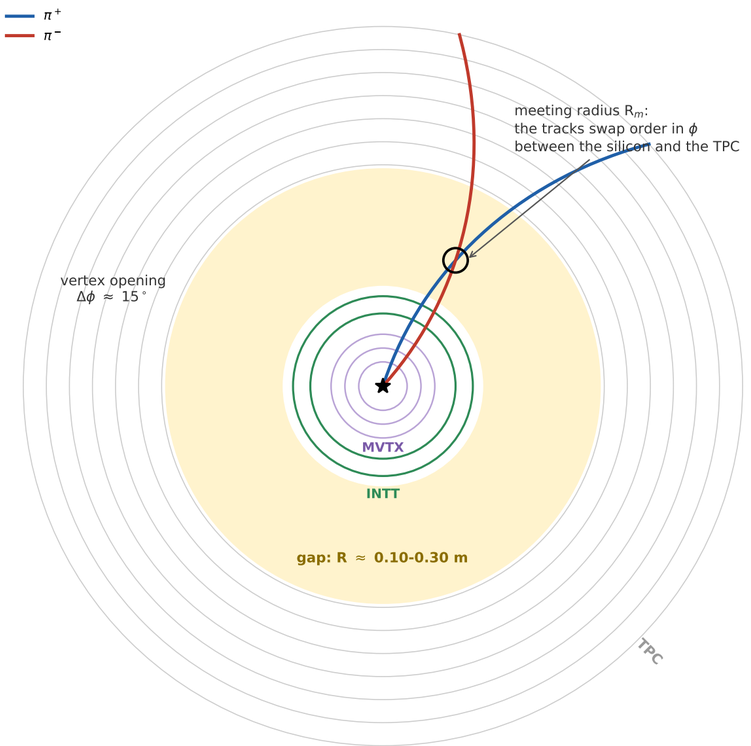


Two real tracks within ~2° at the innermost silicon share or confuse silicon clusters: one of the pair is often lost, like- or unlike-sign, at any dy up to ~1.

ana578 (all of 79516, the loss itself, hlSep): S/M at |dϕ^*(3 cm)| < 1°: LS 0.72, ULS 0.81 at |dy| < 0.2 (ana573 core 0.75-0.85); 1-2° 3-6% low; 2-3° flat.

The cut drops that region of pair space from S and M alike; 2° covers the whole loss region (3° drops loss-free pairs for +0.45%). 6 runs: ULS pions 0.82% S / 0.89% M, LS 0.84% / 0.88% (ana573 0.86/0.94, 0.89/0.93).

15. D1 -- the ±15° ULS notch: two tracks that meet in the INTT-TPC gap
LOST in the silicon-TPC matching -- watched, not removed



Opposite charges bend toward each other; a ~15° vertex opening puts the meeting point at $R_m \approx$ 0.10-0.30 m, where TPC seeds are matched to silicon.

ana578 (all of 79516): ULS S/M 10-20° / 25-50° 0.971 (ana573 0.955-0.975); vs R_m 0.970, 0.941, 0.929 (0.15-0.20 m), 0.932, 0.950, 0.981 = ana573 within 0.01.

The loss is in pairs that MEET inside the gap (0.937) and does not shrink with dy (0.952 at |dy| < 0.1 -> 0.919 at 0.9-1.0): the matching seems to use ϕ only. LS has a shallow notch too. A meeting cut costs 12.9% of the near ULS pairs and leaves ~2%: no cut. A note for tracking.

Panels 13-14: two real particles; the cut drops a region of pair space from sibling and mixed pairs alike (it cannot bring the lost pairs back). Panel 15: a loss upstream of Corral, monitored per production. Numbers: README_SplitTracks578 secs C1, C2, D1.